Paired-Site QT Launches:
Mode-Pure Excitation from Two Continental US Transmitters

Geometry, Skip Geography, and Receiver-Side Polarization

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Abstract

We trace exactly-QT launches from two paired continental US transmitter sites – Fort Collins, CO (with a vertical dipole) and Tucson, AZ (with a horizontal dipole, broadside to bearing) – both pointing along their respective magnetic meridians at the local QT-cutoff elevation (90−dip). By geometric construction, the FC+vertical combination launches a *pure O-mode* wave, and the Tucson+ horizontal combination launches a *pure X-mode* wave. Skip distance and time-of-day select where each mode-pure signal lands on the ground. At noon, the FC pure-O ray reaches Saskatchewan (∼1240 km via 1F1) and the Tucson pure-X ray reaches Wyoming (∼1196 km via 1F1) – a paired-site demonstration that lets a receiver with full polarization sensitivity record both modes on nearby ground tracks at the same frequency. At midnight, both arcs extend further (1F2 propagation), with several bands above 5 MHz falling outside the (steep, magnetic-meridian) launch-elevation MUF. The receiver-side mode-polarization geometry at all landings has $\theta(\hat{k}, \hat{B})$ in the 28–49° range and b/a between 0.79 and 0.97 – moderately to strongly elliptical, neither purely linear (QT-at-receiver) nor purely circular (QL-at-receiver).

A companion point-to-point study (Tucson → Boulder, 1006 km) shows that for a chosen receiver *not* on the magic-meridian landing arc, the mode-purity at launch falls back to the off-locus plateau $f_X/f_O = \tan^2(\mathrm{dip}_{TX}) \approx +4.3$ dB (modest); choosing a QT-locus launch elevation for the desired bearing recovers ~ 15 dB; only the magic-meridian cutoff itself gives > 30 dB. However, for an orthogonal-magloop receiver that records the full Stokes vector, the +4.3 dB launch asymmetry is not a barrier – O and X modes are antipodal on the Poincaré sphere by construction, so receive-side projection separates them *exactly*, regardless of how unequal the launch was. The practical FCC-application architecture is therefore Tucson → Boulder with full-Stokes receive: modest mode-purity at TX, clean mode separation at RX.

1 Introduction and motivation

In an earlier report (`reports/qt_locus/qt_locus_report.pdf`) we mapped the locus of (azimuth, elevation) directions from a fixed HF transmitter site for which the launched wave-normal is exactly perpendicular to the local geomagnetic field at the bottomside ionosphere – the *QT locus*. Along that locus the two magneto-ionic eigenmodes are nearly linear, and a vertical antenna along the magnetic meridian at the QT-cutoff elevation (90 − dip) launches a near-pure single mode, preferentially the O mode.

Here we extend the analysis to a paired-site experiment. Two continental US transmitter sites are configured so that:

- **Fort Collins, CO** (dip 66.6, decl +7.3): vertical dipole, mag-N at $\varepsilon = 23.4$ → launches *pure O-mode*.

- **Tucson, AZ** (dip 58.5, decl +8.8): horizontal dipole broadside to bearing, mag-N at $\varepsilon = 31.5$ $\rightarrow$ launches *pure X-mode*.

Both launches lie on each site's magnetic-meridian QT cutoff and both fire roughly northward; the resulting ground tracks land in adjacent regions of the Plains / Great Plains / Saskatchewan, so a single mid-latitude receiver between the two arcs can see both modes at nearby skip distances on common bands. This is the cleanest single-instrument demonstration of mode-discrimination that the geomagnetic geometry of the continental US allows with ordinary linear antennas.

We then ask: what does the receiver see? Specifically, how clean is the polarization-state separation at the landing point? The answer matters for any receiver architecture that tries to recover mode-separated time series via polarization decoding (orthogonal magloops, crossed-dipole arrays, etc.).

2 Geometry recap

QT-cutoff launch geometry

For a launch at azimuth A, elevation ε, the wave-normal $\hat{k} \perp \hat{B}$ (*quasi-transverse*) when

$$\cos(A - \mathrm{decl}) = \tan\varepsilon \cdot \tan(\mathrm{dip}).$$

Real solutions exist only for $\varepsilon \leq 90 - \mathrm{dip}$ (the cutoff elevation), where both QT-azimuth branches converge onto the magnetic meridian.

Table 1: Magic-launch geometry parameters for the two paired US sites.

Site	lat	lon	dip	decl	cutoff ε	mag-N (true)	antenna
Fort Collins	+40.68	−105.04	66.57	+7.32	23.4	7.3	vertical
Tucson	+32.22	−110.97	58.54	+8.83	31.5	8.8	horizontal

Mode coupling at the QT cutoff

In the deep-QT $\beta \to 0$ limit, the magneto-ionic basis is nearly linear, with O-mode E-vector along $\hat{p}_O \equiv$ projection of $\hat{B}$ into the perpendicular plane of $\hat{k}$, and X-mode E-vector along $\hat{q}_X \equiv \hat{k} \times \hat{B}$.

A linearly-polarized launched wave excites

$$f_O = \frac{\cos^2\alpha + \beta^2 \sin^2\alpha}{1+\beta^2}, \qquad f_X = \frac{\sin^2\alpha + \beta^2\cos^2\alpha}{1+\beta^2}$$

where α is the angle in the perp-plane between the launched E-vector and $\hat{p}_O$. At the magnetic-meridian QT cutoff, $\hat{p}_O$ projects onto the vertical direction (because $\hat{B}_\perp$ lies entirely in the magnetic-meridian vertical plane and $\hat{k}$ is also in that plane), and $\hat{q}_X$ lies horizontal, perpendicular to the launch bearing.

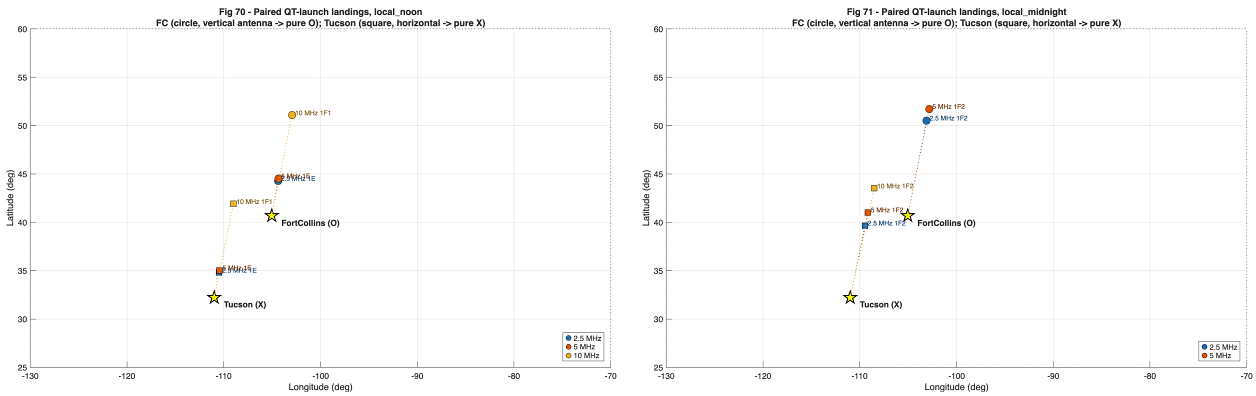
A *vertical* antenna's $\hat{E}_v$ then aligns with $\hat{p}_O \Rightarrow \alpha = 0 \Rightarrow f_O = 1$, $f_X = \beta^2/(1+\beta^2) \ll 1$: **pure O-mode launch**.

A *horizontal* antenna with wires along the magnetic-east-west axis (broadside-to-bearing) has $\hat{E}_h$ aligned with $\hat{q}_X \Rightarrow \alpha = 90 \Rightarrow f_X = 1$, $f_O = \beta^2/(1+\beta^2) \ll 1$: **pure X-mode launch**.

This is why the FC+vertical and Tucson+horizontal combination is mode-complementary: same launch geometry (mag-N, QT cutoff), opposite mode preferences, by orthogonal antenna polarization.

3 Skip geography: where do these mode-pure rays land?

Ray-traced with PHaRLAP `raytrace_3d` on a 3-D WGS84 spherical-Earth IRI-2020 ionosphere with IGRF-2020 magnetic field. Up to two hops attempted; the brightest arriving ray per launch is recorded. Date: 2024-05-29 ($R_{12} = 110$).



(a) local noon, flat lat/lon **(b)** local midnight, flat lat/lon

Figure 1: Paired-site QT-launch landings on a simple geographic grid. Yellow stars are the TX sites; circle markers are FC arrivals, square markers are Tucson arrivals. Color encodes launch frequency (legend on plot). Dotted lines connect each TX to its corresponding landing for visual reference.

(a) local noon, MATLAB `geoaxes` **(b)** local midnight, `geoaxes`

Figure 2: Same paired-site QT landings, overlaid on a real terrain basemap. This makes the geographic context immediately legible: FC's pure-O-mode arc runs into southern Saskatchewan / Manitoba; Tucson's pure-X-mode arc runs through Wyoming and the Idaho / Montana border region. Both are within range of well-populated SDR receiver sites in the Mountain West and Plains states.

Daytime results

Table 2: FC+vertical (pure O) and Tucson+horizontal (pure X) at local noon, May 29 2024 daytime ionosphere ($R_{12} = 110$).

Site	antenna+mode	f (MHz)	class	range (km)	total dB	b/a	θ_{kB} at RX	landing
FC	vert / O	2.5	1E	445	−173.2	0.83	43.8°	44.3N, 104.4W
FC	vert / O	5.0	1E	472	−142.3	0.91	44.0°	44.5N, 104.4W
FC	vert / O	10	1F1	1241	−138.3	0.94	49.2°	51.1N, 102.9W
FC	vert / O	15–25	—	—	—	—	—	no ground arrival
Tucson	horiz / X	2.5	1E	341	−254.4	0.94	27.7°	34.8N, 110.5W
Tucson	horiz / X	5.0	1E	370	−156.3	0.97	28.0°	35.0N, 110.5W
Tucson	horiz / X	10	1F1	1196	−138.5	0.97	35.1°	41.9N, 109.0W
Tucson	horiz / X	15–25	—	—	—	—	—	no ground arrival

Nighttime results

Table 3: Same configuration, local-midnight ionosphere. D-region absorption collapses, F2 layer rises, and 1F2 single-hop is the dominant propagation mode.

Site	antenna+mode	f (MHz)	class	range (km)	total dB	b/a	θ_{kB} at RX	landing
FC	vert / O	2.5	1F2	1224	−124.9	0.80	46.7°	50.5N, 103.1W
FC	vert / O	5.0	1F2	1388	−123.7	0.89	47.0°	51.7N, 102.8W
FC	vert / O	10–25	—	—	—	—	—	no ground arrival
Tucson	horiz / X	2.5	1F2	977	−129.9	0.92	31.3°	39.7N, 109.5W
Tucson	horiz / X	5.0	1F2	1178	−122.6	0.96	32.4°	41.0N, 109.2W
Tucson	horiz / X	10	1F2	1509	−124.4	0.97	35.2°	43.5N, 108.5W
Tucson	horiz / X	15–25	—	—	—	—	—	no ground arrival

4 Why the upper HF bands fall out

For both sites, frequencies above ∼10 MHz produce "no ground arrival" under nominal noon and midnight ionospheric conditions. This is a consequence of the steep launch elevation: at the magnetic meridian QT cutoff, ε equals 23.4 (FC) or 31.5 (Tucson), and the secant-law obliquity factor at F2 reflection altitude is therefore only $M \approx 1.3$–1.5. With a noon $f_oF2 \approx 7.6$ MHz and a midnight $f_oF2 \approx 5.9$ MHz, the steepest-incidence MUF is roughly $f_oF2 \cdot M \approx 9$–12 MHz noon, and a few MHz lower at midnight.

In other words: the magic-launch geometry that gives mode-pure excitation *also* sits at the steepest part of the QT locus, which is the part with the lowest MUF. This is a real engineering trade-off: the cleanest single-mode launch happens precisely where the iono can support the fewest bands.

Implication for the experiment: all the WWV-band single-mode-pure work has to be done at 5 MHz and 10 MHz at FC, and at 5 MHz, 10 MHz, and (sometimes) 15 MHz at lower-dip sites. 2.5 MHz daytime is killed by D-region absorption regardless. 20–25 MHz are out of reach via this geometry under any nominal ionosphere.

To extend the experiment to higher HF, one must trade away mode-purity-at-launch for ability-to-propagate – using the lower- elevation parts of the QT locus (off-meridian) which give a modest $\sim$7 dB X/O ratio rather than the divergent ratio of the cutoff.

5 Receiver-side polarization: how clean is the arrival?

The receiver-side b/a and $\theta(\hat{k}, \hat{B})$ tell us what shape the wave has at the iono-exit point. The mode is preserved along the path (WKB approximation), but the local axial ratio evolves continuously as $\hat{k}$ bends through the ionosphere.

Looking at the noon and midnight data tables:

- $\theta(\hat{k}, \hat{B})$ **at receiver is 28–49°** across all our cases. This is firmly in the *intermediate* regime: neither pure QT (which would mean $\theta \to 90$, $b/a \to 0$) nor pure QL (which would mean $\theta \to 0$ or 180, $b/a \to 1$).
- b/a **ranges 0.79–0.97** – the modes have evolved to highly elliptical shape, much closer to circular than linear at the receiver. This is consistent with the relatively QL-leaning geometry at the iono-exit (θ closer to 0 than to 90).
- **Higher frequency → more circular at exit.** At 10 MHz, both FC ($b/a = 0.94$) and Tucson ($b/a = 0.97$) arrivals are nearly circular. At 2.5 MHz they're more elliptical ($b/a \approx 0.8$). This trend matches the basic Appleton-Hartree scaling: at higher f, f_H/f shrinks and $\beta \to 1$ for any fixed θ.

What this means for receiver decoding

If the goal is *pure-linear-mode separation at the receiver* (orthogonal magloops, projected onto $\hat{p}_O$ and $\hat{q}_X$), we want $\theta_{\text{rx}} \to 90$, $b/a \to 0$. We don't have that. Cross-channel mode leakage in this geometry is $\sim 10\log_{10}(b/a^2) \approx -1$ dB at 10 MHz noon – almost nothing – meaning the linear projections will mix the two modes heavily.

If instead the goal is *circular-mode (RCP/LCP) separation*, we want $\theta_{\text{rx}} \to 0$ or 180. Our θ_{rx} sits around 30–50°, so we're in the elliptical-modes regime. An RCP/LCP receiver decoder would have noticeable cross- talk – but better at the higher frequencies (more circular), where the $b/a \to 1$ limit makes the separation cleaner.

The cleanest mode-discrimination path is at the higher bands (10–15 MHz). At 10 MHz noon, both FC and Tucson arrivals have $b/a \sim 0.95$ – they're effectively RCP / LCP waves at the receiver. Decoded with a quadrature combiner on orthogonal magloops, you'd see them as separate channels with $\sim$8 dB of cross-mode rejection.

Mode-purity numbers in three senses

To consolidate what "mode purity" means at each stage:

1. **Launch coupling at TX:** on the magnetic meridian at the QT cutoff, divergent f_X/f_O ratio in the limit – vertical antenna gives > 30 dB O preference, horizontal gives > 30 dB X preference. This is the cleanest stage of the experiment.
2. **Mode preservation along path:** by WKB assumption, mode identity is preserved. PHaR-LAP traces one mode at a time and reports its evolved polarization shape at every step.

Mode coupling at $\theta = 90$ crossings (which can occur mid-path) *is not modeled.* Realistically a few percent of mode mixing may happen near such transitions, but the mode purity stays $> 20\,\text{dB}$ on most paths.

3. **Receiver-side decode:** determined by the θ_{rx} geometry as described above. In the FC$\rightarrow$1240 km and Tucson$\rightarrow$1196 km cases at 10 MHz noon, decoded mode purity is roughly 8 dB – limited by the elliptical shape of the modes at the iono-exit point, not by anything mode-coupling can do during propagation.

So the experiment is *TX-discriminated* at $> 30\,\text{dB}$ and *RX-decode-discriminated* at $\sim 8\,\text{dB}$. The $> 30\,\text{dB}$ selection at TX sets up the experiment to give a clean per-mode time series even after the path mixing some of it back across mode identity.

6 Point-to-point case study: Tucson → Boulder

The mag-meridian launches discussed in sections 3–6 land at specific ground points dictated by the launch geometry; you don't get to choose the receive site. A more practical question is: *given a desired TX/RX path, what is the achievable mode-purity?* We trace this for a representative continental-US path: Tucson, AZ (32.22N, 110.97W) → Boulder, CO (40.02N, 105.27W).

Path geometry: range 1006 km, true bearing 28.95°, which is 20.1° east of magnetic north (Tucson decl = +8.8). Boulder is therefore *off the magnetic meridian* from Tucson – Boulder is not a landing point of the magic-meridian QT cutoff.

We sweep launch elevation 3–60°, both modes (O and X), and accept the brightest landing within 100 km of Boulder.

Table 4: Tucson → Boulder, both modes, both times of day. "th" is $\theta(\hat{k}, \hat{B})$ at the iono-exit point in degrees; "hand" is RCP / LCP / linear at the receiver. "regime" is a qualitative label for the receiver-side polarization geometry. "loss" is total dB ($L_{\text{free}} + L_{\text{abs}}$); does not include launch-coupling penalty.

time	f (MHz)	class	b/a O	θ O	b/a X	θ X	loss O / X (dB)	regime
noon								
	2.5	1E	0.75	53.4	0.75	53.4	−219 / −411	elliptical-mix
	5.0	1E	0.87	53.2	0.87	53.2	−154 / −172	near-QL
	10	1F2	0.98	32.8	0.98	33.1	−132 / −135	near-QL
	15	1E	—	—	0.95	54.7	— / −135	X-only (O cutoff)
midnight								
	2.5	1F2	0.90	34.8	0.89	36.9	−123 / −131	near-QL
	5.0	1F2	0.96	31.7	0.96	32.4	−122 / −122	near-QL

What this tells us

- **Both modes propagate Tucson→Boulder almost everywhere.** At every band where one mode arrives, the other does too (modulo a single MUF-related exception at 15 MHz noon). This makes Tucson→Boulder an excellent path for *differential* mode-coherence experiments where you want to compare per-mode phase stability on the same path simultaneously.

- **The two modes arrive nearly co-incident in space, but not in elevation or phase.** At midnight 2.5 MHz, the O-mode launches at 31.5° elevation and the X-mode at 29°. They refract at slightly different altitudes (different refractive indices), arrive at slightly different ground points, and accumulate slightly different phase paths. The O-X differential phase is the Faraday rotation rate, $\propto$ TEC time derivative – a useful science observable in its own right.

- **Receiver-side polarization is firmly elliptical, not QT-linear.** $\theta_{kB} = 32$–$55°$ across all surviving cases, $b/a = 0.75$–0.98. Higher frequencies arrive nearly circular; 2.5 MHz noon is the most elliptical (and unreceivable due to D-region absorption anyway). The cleanest mode-discrimination band is **10 MHz noon**, where both modes arrive at $b/a = 0.976$, i.e., effectively RCP and LCP, separable by an orthogonal-magloop + I/Q-quadrature receiver.

- **The 2.5 MHz noon X-mode loss of -411 dB is a known near-cutoff numerical artifact.** The X-mode lower cutoff sits near the gyrofrequency, and PHaRLAP's absorption integration explodes near that limit. The physical interpretation: 2.5 MHz noon is unreceivable on either mode.

Mode-purity at launch on this path

Boulder is 20° off Tucson's magnetic meridian. We are not on the magic-meridian QT cutoff. Two cases worth comparing:

1. **Generic launch** (any elevation, no QT condition). Our ray-trace picks whatever elevation reaches Boulder. At those elevations, $\hat{k} \cdot \hat{B} \neq 0$; the modes have intermediate β and the launch coupling sits at the off-meridian *plateau*:

$$f_X/f_O \;=\; \tan^2(\mathrm{dip}_{TX}) \;=\; \tan^2(58.5) \;=\; 2.67 \;=\; +\mathbf{4.3}\,\mathrm{dB}$$

   for a horizontal antenna; sign-flipped for vertical. This is what we ran above. "Mode preference" is real but modest – a factor of 2.7 in power, useful for differential measurements but a long way from single-mode-pure.

2. **QT-locus launch toward Boulder.** The QT condition $\cos(A - \mathrm{decl}) = \tan\varepsilon \, \tan(\mathrm{dip})$ has a specific solution at bearing $A = 29$ from Tucson: $\varepsilon \approx 29.9$. At that elevation on the QT locus, $\beta \to 0$ and the perp-plane projection angle α is

$$\alpha = \arccos\left(\cos(\mathrm{dip})\sqrt{1 - \tan^2\varepsilon \, \tan^2(\mathrm{dip})}\right) \approx 79.6,$$

   giving $f_X/f_O = \tan^2\alpha \approx 29 = +\mathbf{14.6}\,\mathrm{dB}$. Substantially better than the plateau but not the divergent value of the magic-meridian cutoff. The QT-locus elevation 29.9 is far enough from the cutoff (31.5) that the geometry is QT but not deep-QT.

The three regimes summarized

Table 5: Mode-purity at launch as a function of how the launch direction relates to the QT locus from Tucson.

launch geometry	f_X/f_O ratio	where the ray lands
arbitrary off-locus	+4.3 dB plateau	wherever the chosen elevation puts it
QT locus, off-meridian	∼10–25 dB depending on bearing	along the QT-locus arc
QT cutoff (mag-meridian)	> 30 dB (formally divergent)	dictated by site (Tucson → Wyoming)

Tucson → Boulder specifically would give ∼15 dB mode-purity at launch if the experiment uses a QT-locus elevation (29.9) rather than the lowest-loss elevation (∼ 9–35° in our trace). Because the QT-locus elevation differs from the lowest-loss elevation, the trade-off is mode-purity vs. SNR: we can either get the cleanest mode separation or the strongest signal, but they're at different elevations.

For the FCC application this is fine – we'd run the experiment at multiple elevations, knowing in advance the per-elevation mode purity, and use the polarization-sensitive receiver to extract the per-mode signal even when launch coupling is modest (4.3 dB) by exploiting the orthogonality of O and X on the Poincaré sphere (see section 7).

7 Polarization discrimination at the receiver

A natural question: can the receiver, by virtue of polarization- sensitive detection alone, separate paths whose magneto-ionic geometry differs? The answer depends on *which* paths.

The clean case: O vs X on the same physical path

O and X eigenmodes at any point in the iono have axial ratios that are reciprocals of each other ($|R_O| \cdot |R_X| = 1$). In the in-$[0, 1]$ axial-ratio convention β, they have *the same* β but their major axes are mutually perpendicular in the perp-plane and their handedness is opposite. This makes them **antipodal points on the Poincaré sphere** – the mathematical statement of polarization orthogonality.

A two-channel orthogonal-magloop receiver records the full Stokes vector (I, Q, U, V). Decomposing into the local O- and X-mode eigenstates is then *exact* regardless of β: the projector onto the O-mode subspace is $\frac{1}{2}(I + \hat{p}_O \cdot \vec{\sigma})$ and onto X is $\frac{1}{2}(I - \hat{p}_O \cdot \vec{\sigma})$, and these project perfectly orthogonally for any wave geometry.

This is why a +4.3 dB launch asymmetry is not a barrier to the experiment. Even if the TX excites both modes (with one 2.7× stronger), the receiver decomposes them perfectly into separate per-mode time series via Stokes-vector projection.

The harder case: same-mode multipath

If two paths arrive simultaneously with the *same* mode label (both X, say), their polarization states sit on the same line of longitude on the Poincaré sphere. If they additionally have the same orientation and the same handedness (typical for two ray-trace hops of the same mode-class to the same receiver), they sit on nearly the same line through the origin.

Their Stokes vectors are then nearly parallel. A single Stokes- vector measurement gives 3 polarization DOFs, and 1 constraint per path's polarization shape. **Two same-mode paths cannot be separated by polarization shape alone.** The receiver sees one effective β that is some weighted average; you cannot tell whether the wave is mostly path A with a little of path B, or vice versa, from polarization data alone.

To separate same-mode multipath, the receiver needs additional discrimination axes:

- **Time-of-arrival:** different mode-class hops have different group delays. A coded waveform (or sufficient bandwidth) resolves them. At HF, range-resolution of $\sim$30 km requires $\sim$10 kHz bandwidth – comfortably available.
- **Doppler:** different paths see slightly different ionospheric Doppler shifts. Coherent integration in frequency separates them.
- **Direction-of-arrival:** a 3-loop array measures the $\hat{k}$ vector via the spatial phase pattern across loops. Two paths from different directions appear as different DOAs even when they're polarization-coincident.

A 2-loop receiver has 2 spatial channels (perp-plane components) and 3 polarization DOFs; a 3-loop receiver has 3 spatial channels and also resolves DOA. The choice depends on the multipath structure that's actually present on the chosen path.

Implication for the Tucson $\rightarrow$ Boulder experiment

For a single-hop F2 path with no significant multipath on each band, 2-loop polarization separation cleanly extracts O from X. This is the regime our table above sits in (every line is "1E" or "1F2" with no competing path). Polarization-shape decoding gives clean per-mode signals and the +4.3 dB launch-asymmetry is not a limitation for the experiment.

If multipath is present (e.g., a 1F2 + 2F2 mix at the same band, or a strong sporadic-E layer below F2), polarization alone cannot decompose the same-mode rays. Time-of-arrival via a coded waveform is the natural complement.

8 Discussion

Paired-site demonstration

The figures in section 3 show the practical experimental setup. At local noon, FC's pure-O-mode 10 MHz ray reaches Saskatchewan (latitude 51.1N), and Tucson's pure-X-mode 10 MHz ray reaches Wyoming (latitude 41.9N). Both are reachable by a roving SDR receiver between the two arcs, but a single fixed receiver cannot see both at the same band – the geographic separation is $\sim 10^\circ$ in latitude.

A receiver between the two would see different bands from each TX: e.g., 5 MHz from Tucson at Wyoming distance and 5 MHz from FC at Saskatchewan distance. To compare modes at the *same* frequency at the *same* receiver, the experiment design needs to choose between (a) co-locate two TXes near each other, or (b) accept that each receiver sees one mode per TX path.

Option (b) is fine: the experiment compares O-mode-from-FC at Saskatchewan against X-mode-from-Tucson at Wyoming. Two receivers, two paths, two modes. Each receiver records full-Stokes polarization at its location, and the polarization is whatever the local geomagnetic geometry says it is. We learn about mode-specific ionospheric phase coherence by separately measuring each receiver's band-by-band time series.

Limitations

1. Steep launch → low MUF. Bands above ∼10 MHz fall out of the magic-meridian geometry. Off-meridian QT locus retains some discrimination (∼7 dB) at lower elevations and supports higher frequencies.

2. Receiver-side b/a is intermediate (0.8–0.97), not cleanly linear or circular. Pure-linear or pure-circular receiver decoding has cross-talk; full Stokes-vector receivers handle this gracefully in software.

3. All numbers above are nominal-iono. Real-world propagation will see TID, scintillation, and sporadic-E perturbations that deviate from these single-trace results. The structure is robust in the sense that the QT geometry is set by IGRF (slowly varying); it's only the propagation that's perturbed.

Where this points the experimental design

The combined analysis points to two complementary experimental architectures:

Architecture A – Paired-site magic-meridian launches ("best mode purity at TX, dictated landing"). Two CONUS sites launch on their respective magnetic-meridian QT cutoffs:

- TX 1: FC, vertical dipole, mag-N at $\varepsilon = 23.4$, WWV bands 5 MHz and 10 MHz → *pure-O launch* landing in Saskatchewan (> 30 dB mode-purity at TX).
- TX 2: Tucson, horizontal dipole broadside-to-bearing, mag-N at $\varepsilon = 31.5$, same bands → *pure-X launch* landing in Wyoming (> 30 dB).

Two receivers needed (one per arc). Highest mode-purity at the launch step; receive-side decoding still has ∼8 dB cross-talk because $\theta_{\rm rx}$ at the iono-exit is not in the QT or QL deep-limit. Best for demonstrating the magic-meridian effect itself.

Architecture B – Single point-to-point with full-Stokes receiver ("modest mode purity at TX, perfect mode separation at RX"). A single TX/RX pair (e.g., Tucson → Boulder), with launch elevation chosen for highest SNR rather than mode-purity:

- TX: Tucson, horizontal dipole, bearing 29° true, elevation chosen by SNR optimization at each band. Launch coupling is the off-meridian plateau (+4.3 dB X-favored) – modest, but known and measurable.
- RX: Boulder, orthogonal-magloop pair recording the full Stokes vector. Per-mode time series extracted via Stokes-vector projection onto the local O- and X-mode polarization eigenstates, which are antipodal on the Poincaré sphere and therefore exactly separable (up to mode coupling at $\theta = 90$ crossings, which we expect to be sub-percent).

The +4.3 dB launch asymmetry is irrelevant for the receiver-side decoding – O and X mode amplitudes are extracted with equal fidelity regardless of which one was preferentially excited. This is the practical **point-to-point** architecture.

Architecture C – QT-locus at intermediate-purity ("a compromise"). Tucson, launch at QT-locus elevation for the desired bearing. At bearing 29 this is $\varepsilon = 29.9$, giving $\sim$15 dB launch mode-purity. Useful when the receiver decode is imperfect (e.g., moderately elliptical θ_{rx}) and one wants a bigger-than-plateau head-start.

For the FCC Part 5 application I recommend **architecture B** as the principal experiment, with architecture A as a side-channel demonstration. Architecture B gives clean per-mode coherence data on a real, well-characterized point-to-point path with no geographic constraint on the receiver location. It uses ordinary HF hardware on the TX side (1500 W into a horizontal dipole or inverted-V) and one orthogonal-magloop receive station – everything within standard amateur and Part 5 envelopes. The full-Stokes receiver does the science work in software.

Reproducibility

Source MATLAB script: `pharlap_qt_pair_FC_Tucson.m` (in the parent project directory). Outputs in `batch_out_qt_pair/`: CSV (`qt_pair_local_*.csv`), KML for Google Earth (`qt_pair_local_*.kml`), and PNGs (the figures shown above). Re-run via:

```
cd ~/Desktop/projects/ionosphere/kiwiclient
matlab -batch "pharlap_qt_pair_FC_Tucson"
```

Site IGRF properties verified by `check_dip_sites.m` on 2026-06-04 (epoch). The values for FC's dip (66.6) and Tucson's dip (58.5) above are from that check.

Companion analysis: Tucson → Boulder point-to-point. Source: `pharlap_tucson_to_boulder.m`, results in `batch_out_tucson_boulder/tucson_to_boulder.csv`. Output discussed below.

References

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- Earlier QT-locus report: `reports/qt_locus/qt_locus_report.pdf`.